SIMPLEST FORM

Rationalizing with the conjugate

Multiplying by the conjugate makes a difference of squares below the bar, and the radical leaves the denominator.

01

BINOMIAL DENOMINATOR — BEFORE

$$\frac{3}{2+\sqrt{5}}$$

↓ Multiply by the conjugate — (2 + √5)(2 − √5) = 4 − 5 = −1

02

DENOMINATOR IS -1

$$\frac{3(2-\sqrt{5})}{-1}$$

↓ Clear the -1 — every sign flips: −3(2 − √5)

03

RATIONAL DENOMINATOR

AFTER

$$-6+3\sqrt{5}$$

▼ CHECK THE SIGN